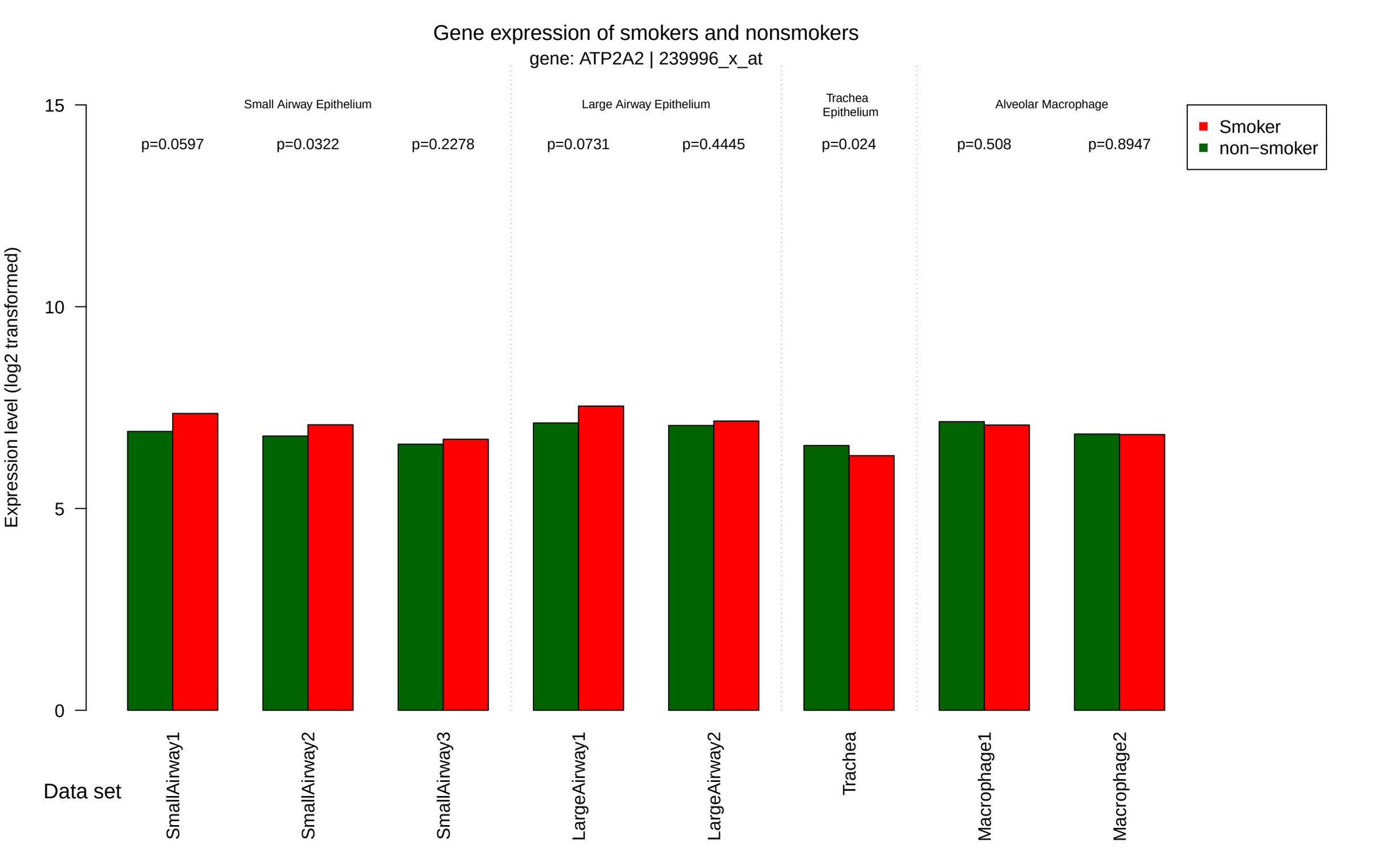
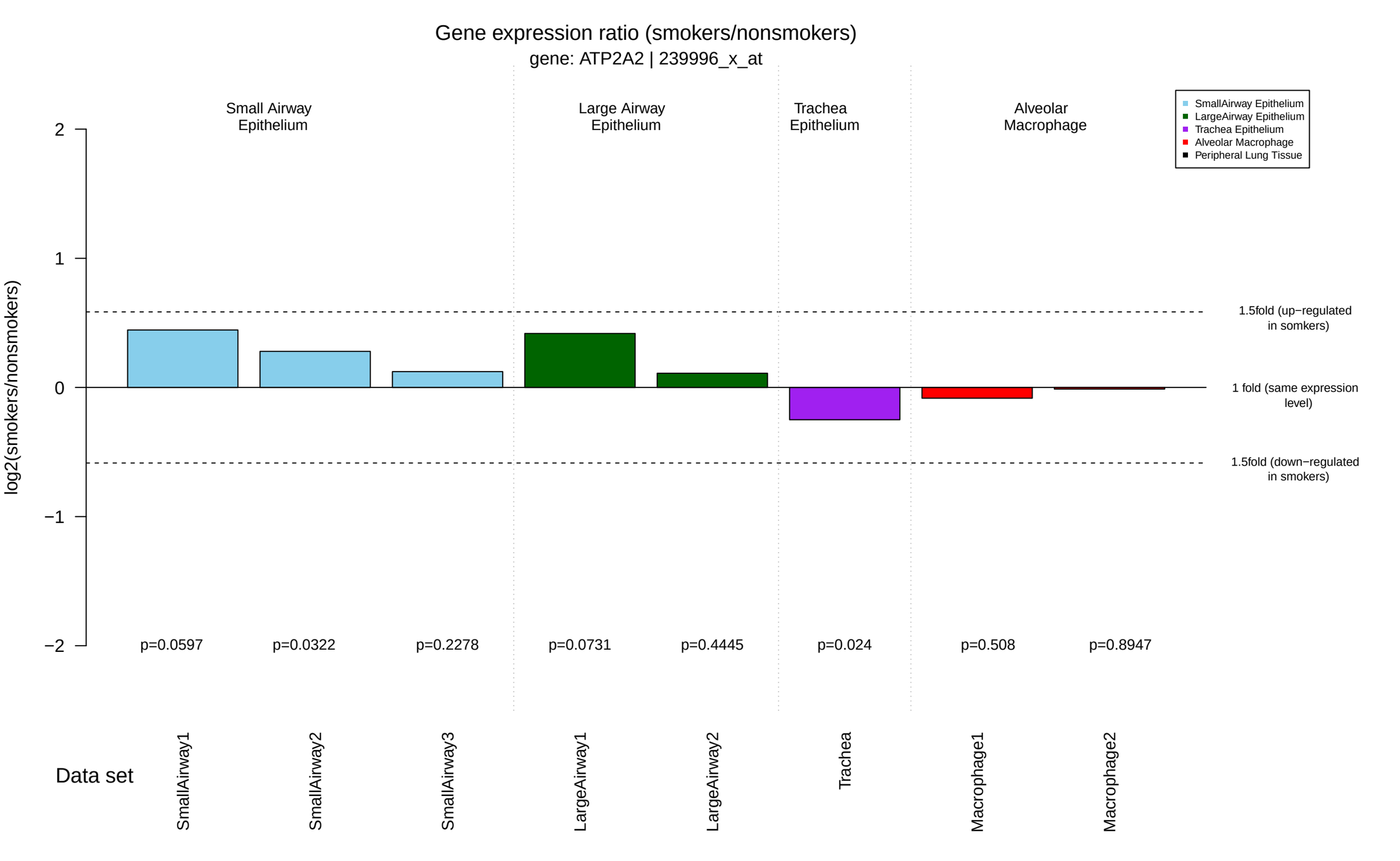
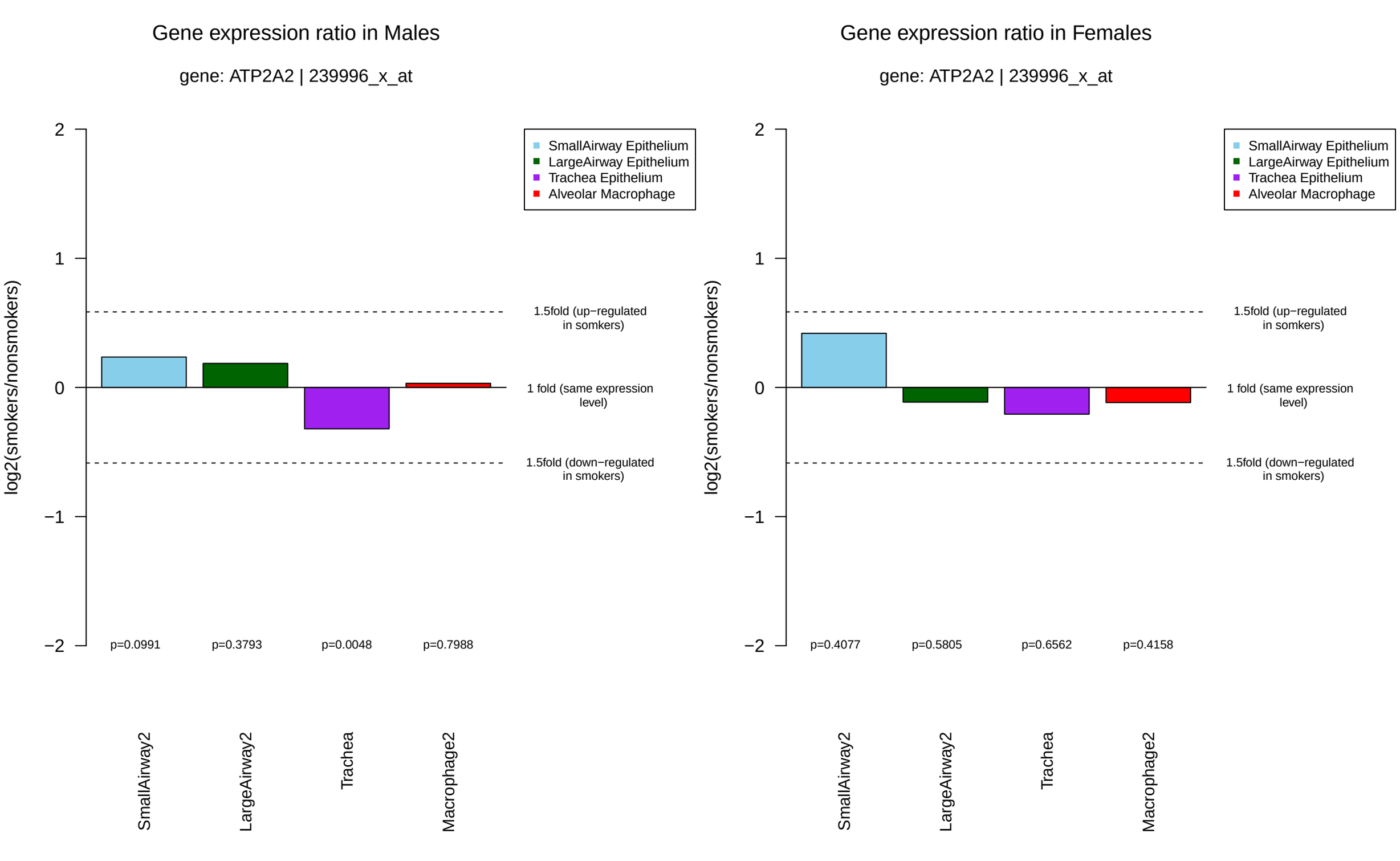
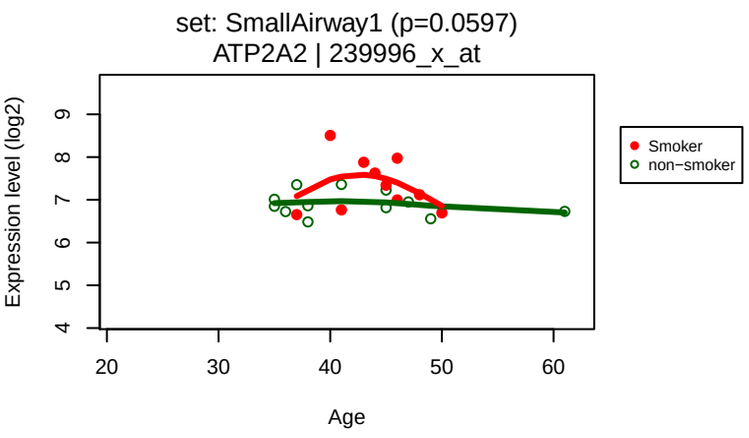
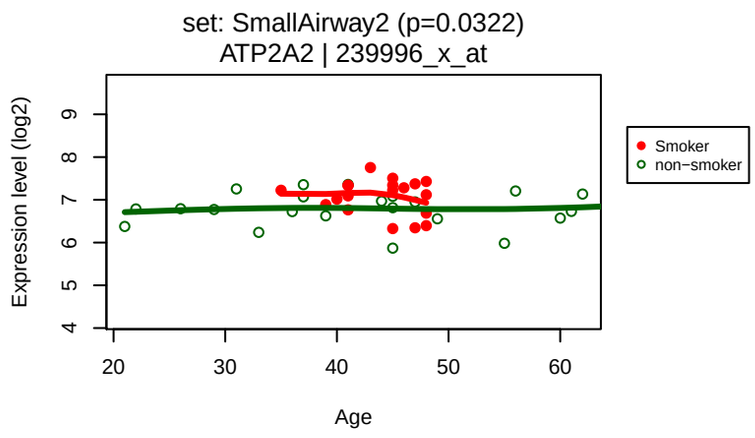
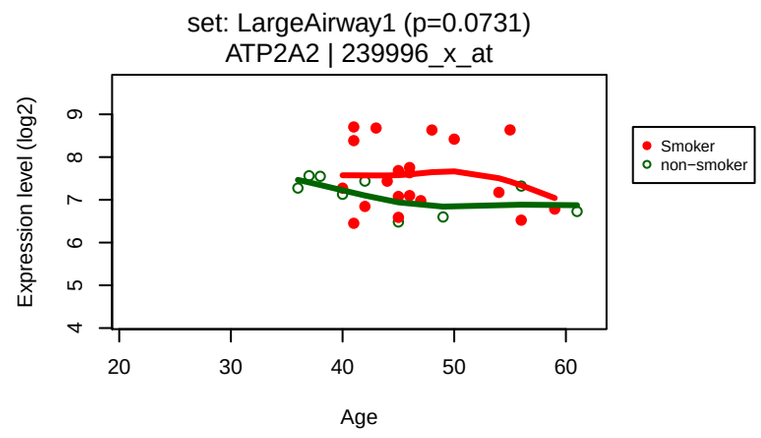

Gene expression of smokers and nonsmokers
gene: ATP2A2 | 239996_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0597
p=0.0322
p=0.2278
p=0.0731
p=0.4445
p=0.024
p=0.508
p=0.8947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

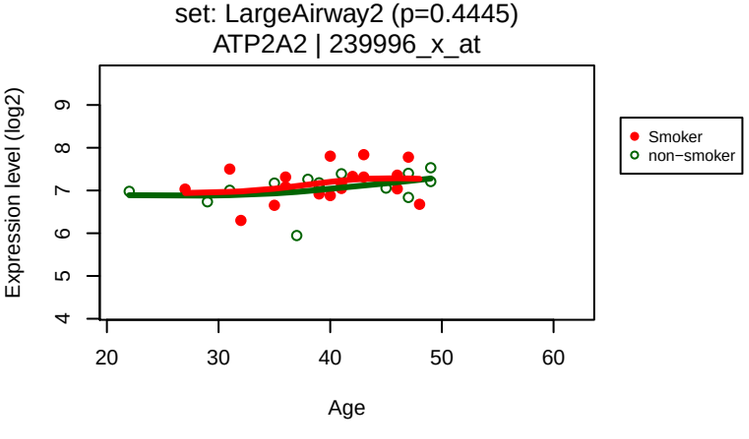

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A2 | 239996_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0597
p=0.0322
p=0.2278
p=0.0731
p=0.4445
p=0.024
p=0.508
p=0.8947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

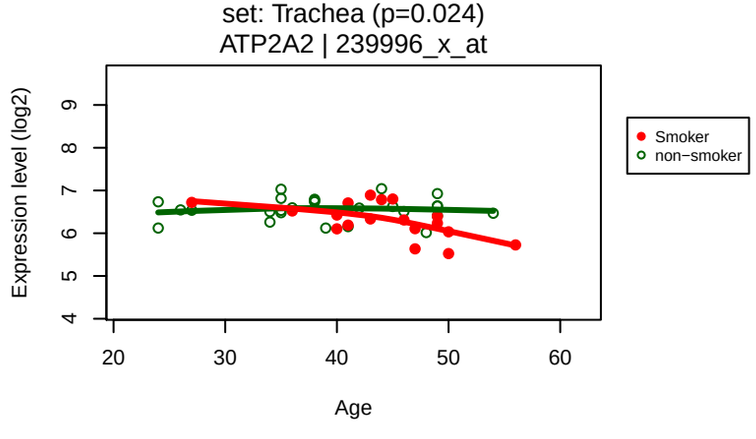

Gene expression ratio in Males
gene: ATP2A2 | 239996_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0991
p=0.3793
p=0.0048
p=0.7988
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A2 | 239996_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4077
p=0.5805
p=0.6562
p=0.4158
SmallAirway2
LargeAirway2
Trachea
Macrophage2

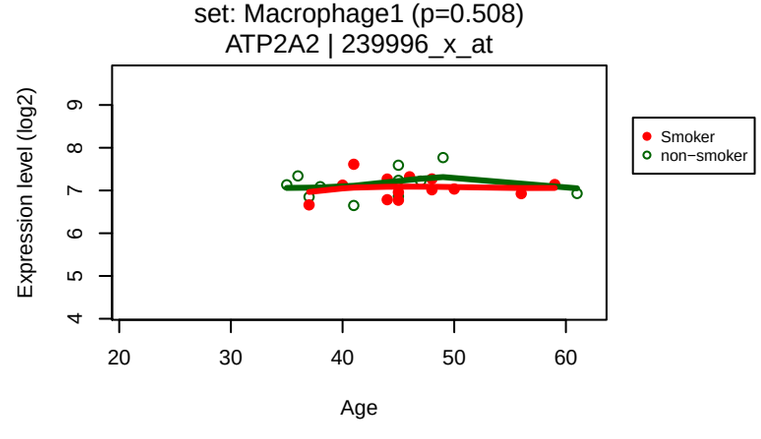

set: SmallAirway1 (p=0.0597)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker

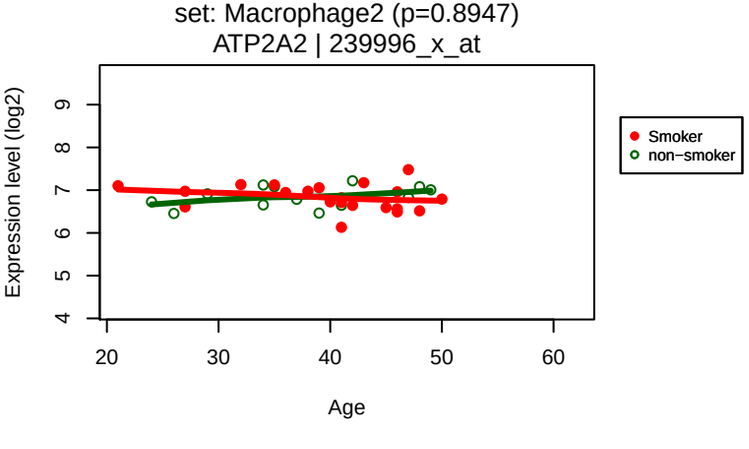

set: SmallAirway2 (p=0.0322)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker

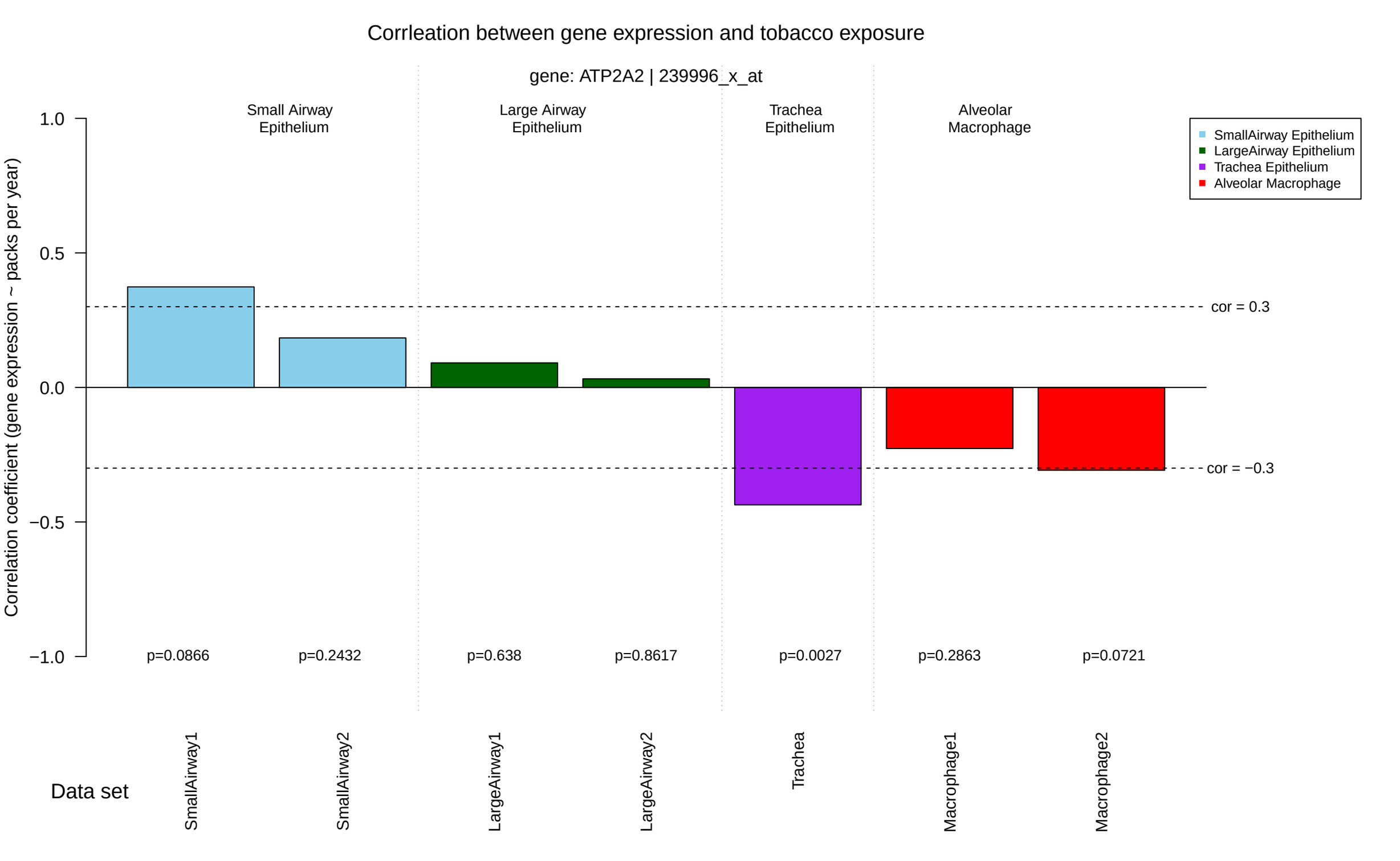

set: LargeAirway1 (p=0.0731)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4445)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.024)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.508)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8947)
ATP2A2 | 239996_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A2 | 239996_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0866
p=0.2432
p=0.638
p=0.8617
p=0.0027
p=0.2863
p=0.0721
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2